

Employee Workforce Data Cleaning & Analysis

A full pipeline from raw, messy records to actionable HR insights

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Executive Summary



This engagement transformed a deliberately messy, inconsistent employee dataset into a clean, analysis-ready resource — then extracted meaningful workforce insights on performance, compensation, remote work, and hiring trends.

1,020

Employee records processed

13

Visualizations produced

5

Phases of workflow

Project Scope & Methodology

The engagement followed a standard, industry-proven analytics workflow using Python (pandas, matplotlib, seaborn) and PostgreSQL.



Data Cleaning: Key Steps

Missing Age Values (211 records)

Filled with median age of 30 years — preserving all 1,020 records.

Missing Salary Values (24 records)

Filled using group-aware median by department and region.

Department-Region Split

Combined field (e.g., "DevOps-California") split into two independent columns.

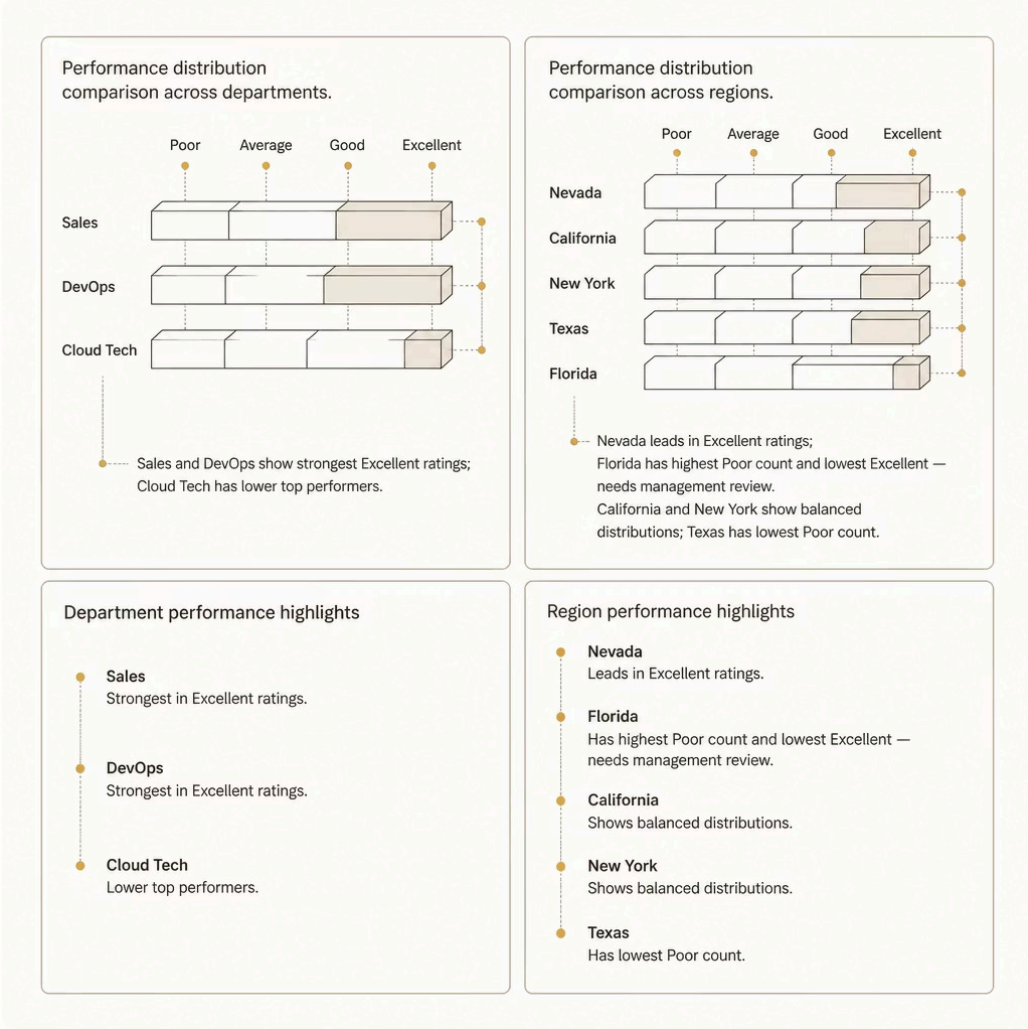
PII Removal & Name Consolidation

Email and phone removed; first/last names merged into a single employee_name field.

Performance Score Standardization

Text categories mapped to numeric scale (1–4) for calculations, then restored to labels for reporting.

Performance Distribution by Department & Region



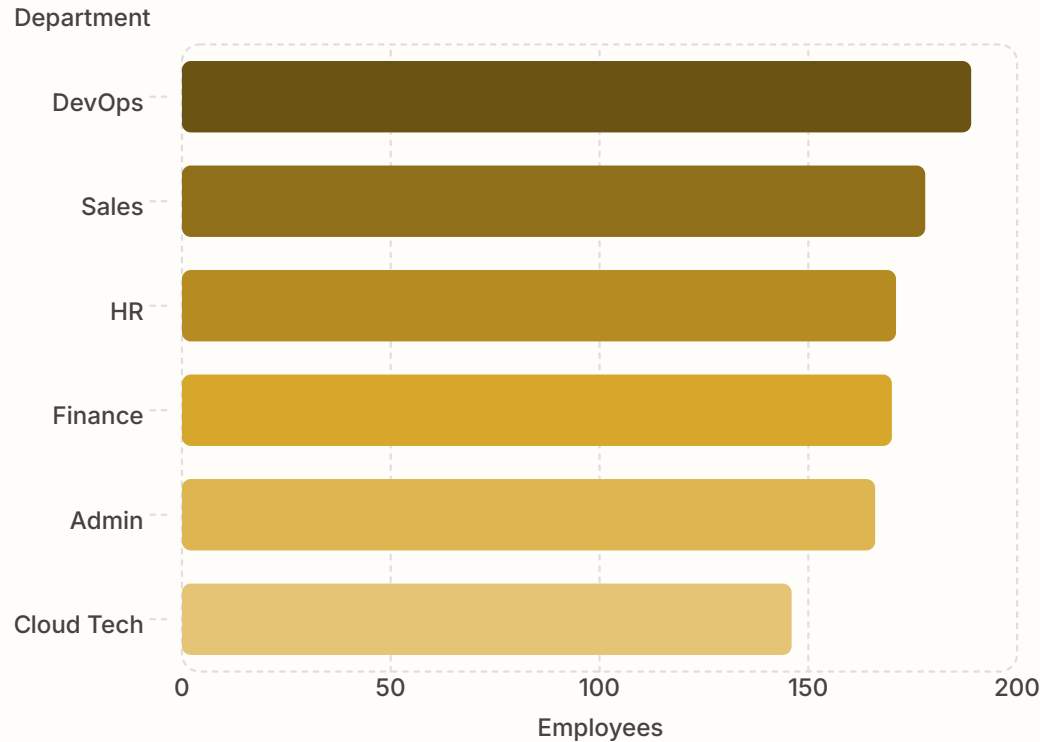
Department Highlights

Sales and DevOps show the strongest "Excellent" ratings. Cloud Tech has fewer top performers. HR has the widest performance spread.

Regional Highlights

Nevada leads in "Excellent" performers. Florida has the highest "Poor" count — warranting management review. Texas has the lowest "Poor" count.

Employee Distribution & Demographics



By Region

California (187) and Florida (185) are the largest regions. Texas is smallest (153). Regional spread is balanced — only 18% gap between largest and smallest.

By Status

Pending (356) is the largest status group, just ahead of Active (352). Inactive: 312. The high Pending count may signal onboarding bottlenecks.

By Age Group

Performance is evenly spread across Young-Adult (2.60), Adult (2.53), and Middle-Aged (2.54) — age does not drive performance outcomes.

Remote Work Patterns



Remote work adoption varies widely by location and department — no single company-wide pattern.

Most Remote

Florida DevOps: 24 remote vs. 10 onsite

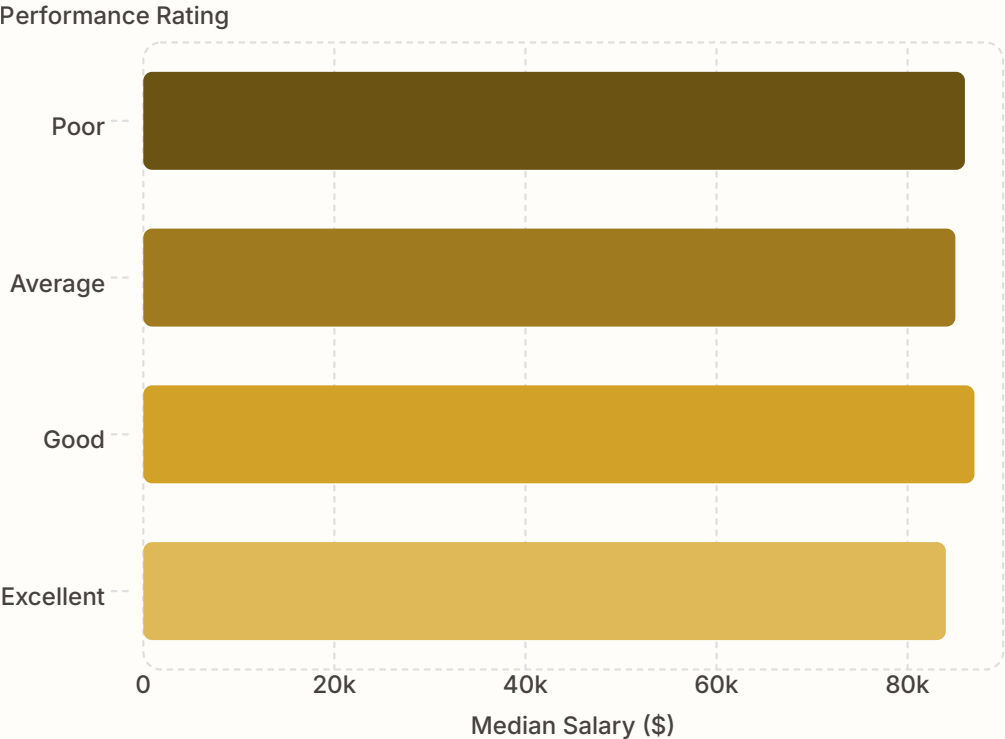
Most Onsite

New York Cloud Tech: 5 remote vs. 19 onsite

Performance Parity

Remote and onsite performance distributions are nearly identical — no evidence that work mode affects outcomes

Salary Insights



Pay Does Not Track Performance

Median salaries across all four performance tiers fall within \$84K–\$87K. "Excellent" performers do not earn meaningfully more than "Poor" performers — compensation appears driven by role, department, or tenure rather than individual performance.

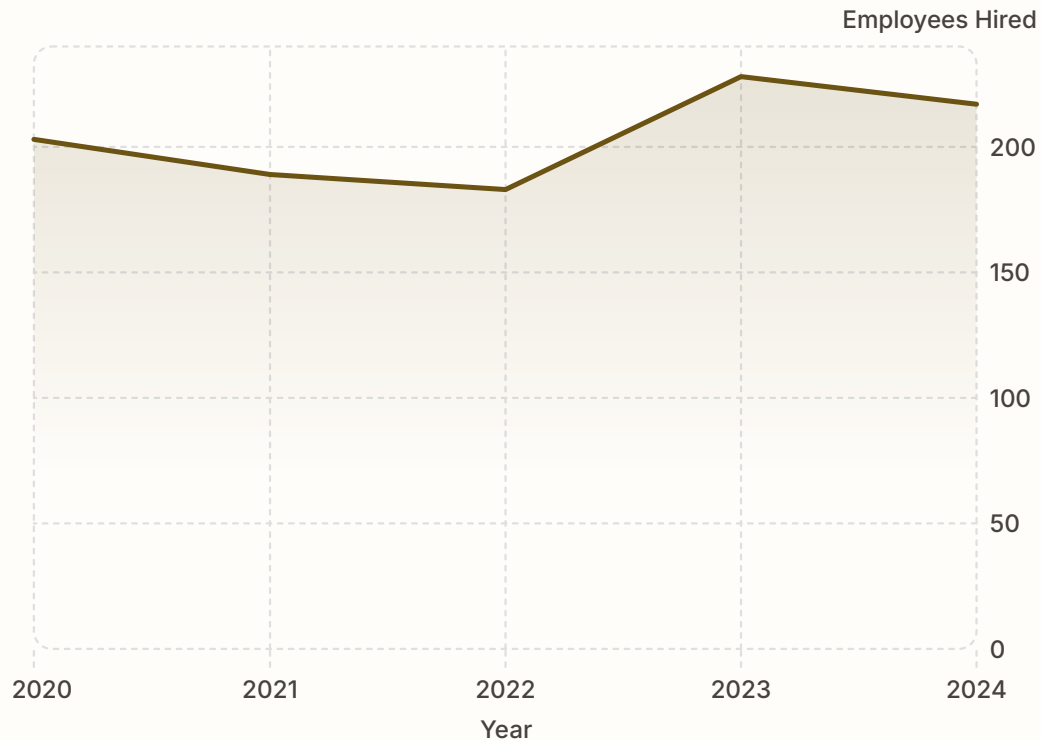
Top Pay by Department × Region

Admin in Illinois (\$92,417) and Cloud Tech in Florida (\$92,126) are the highest-paid combinations. Finance in Illinois and Nevada are the lowest (~\$75,700–\$76,000).

Remote vs. Onsite Pay

Near-identical salary distributions — median \$86,500 (onsite) vs. \$85,000 (remote). Work mode is not a major pay driver.

Hiring Trend & Overall Performance



Hiring Trend

Hiring dipped from 203 (2020) to a low of 183 in 2022, then surged to 228 in 2023 — the highest year — before easing to 217 in 2024. This pattern suggests recovery from a constrained period.

Performance Distribution

Ratings are evenly spread: Good (270), Average (267), Excellent (267), Poor (216). The balanced distribution suggests a credible performance evaluation process.

Key Findings & Recommendations

1 Investigate pay-performance disconnect

Salary does not track performance rating — consider a formal compensation review.

2 Review Florida's performance profile

Highest "Poor" rating count of any region — management review warranted.

3 Examine the Pending status backlog

356 employees in Pending status — confirm whether this reflects onboarding bottlenecks.

4 Use heatmaps for workforce planning

Department × Region concentrations (e.g., HR in Florida) can inform resourcing decisions.

5 Maintain PostgreSQL as single source of truth

Future reporting should build on the cleaned employee_details table.